

Circular Motion Fundamentals:

Physical Parameters associated with the circular motion are

- 1) Angular Displacement (θ) rad
- 2) Angular Velocity (ω) rad/s
- 3) Angular Acceleration (α) rad/s²

Relation b/w them is also same as Rectilinear Motion

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \omega \cdot \frac{d\omega}{d\theta}$$

Circular motion is defined only for point masses or when size of object $\ll$ Radius of the circle.



Motion of Earth around Sun can be called circular motion bcz

[Size of Earth $\ll$ R]

$$v = R\omega$$

Otherwise, if continuous body/size of body comparable with radius, then it becomes 'Rotational Motion'

Acceleration

Tangential (a_t)

$$\frac{dv}{dt} = R \frac{d\omega}{dt}$$

$a_t = R\alpha$
Responsible for change in magnitude

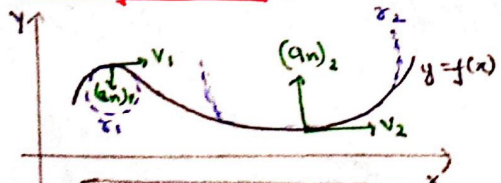
(a_n) Normal/Centripetal

$$a_n = \frac{v^2}{R} = \omega^2 R$$

Responsible for change in dirⁿ.

$$a_{total} = \sqrt{a_t^2 + a_n^2}$$

Radius of Curvature:



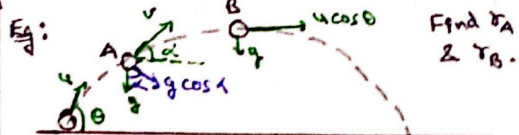
Curvature $\propto \frac{1}{\text{radius of curvature}}$

Smaller the radius, steeper the curve.

At any point (x_1, y_1), radius of curvature is given by

$$r_c = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|d^2y/dx^2\right|} \quad x_1, y_1$$

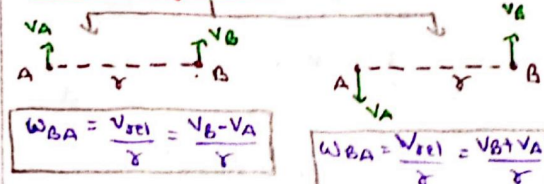
Circular Motion



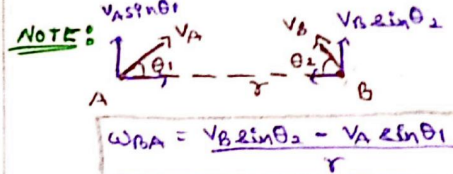
$$r_A = \frac{v^2}{a_n} = \frac{u^2 \cos^2 \theta}{\cos^2 \theta} \times \frac{1}{g \cos \theta} \Rightarrow \frac{u^2 \cos^2 \theta}{g \cos^3 \theta}$$

$$r_B = \frac{v^2}{a_n} = \frac{u^2 \cos^2 \theta}{g}$$

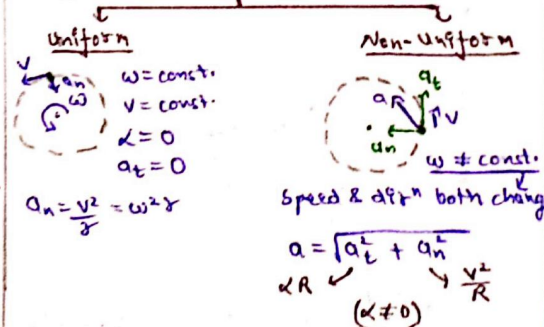
Relative Angular Velocity:



(v_A ko point ke vectorially add kr diya)



Uniform v/s Non-Uniform Circular Motion:



Non uniform Motion when $\alpha = \text{const.}$:

In case of uniform α , all the vector eqⁿs of motions are applicable.

- $\omega_f = \omega_i + \alpha t$
- $\omega_f^2 - \omega_i^2 = 2\alpha\theta$
- $\theta = \omega_i t + \frac{1}{2}\alpha t^2$
- $\Delta\theta = \omega_i t + \frac{\alpha}{2}(2\pi n)$

NOTE:

- 1) If both ω & α are in same sense (CW or ACW), then take $\alpha \rightarrow +ve$
- 2) If α & ω are in opp. sense, then take $\alpha \rightarrow -ve$

Convert rpm into rad/s always

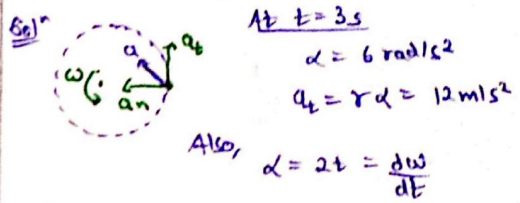
$$x \text{ rpm} \rightarrow \frac{x \times 2\pi}{60} \text{ rad/s}$$

Non-uniform Circular Motion when $\alpha \neq \text{const.}$

Use differential relations of circular parameters.

$$\omega = \frac{d\theta}{dt} \quad \alpha = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta}$$

Ex: $\alpha = 2t$ & particle starts from rest. At $t = 3s$, find a . ($r = 2m$)



$$\int_0^{\omega} d\omega = \int_0^{3t} 2t \cdot dt \Rightarrow \omega = 9 \text{ rad/s}$$

$$a_n = \omega^2 r = 81 \times 2 \Rightarrow 162 \text{ m/s}^2$$

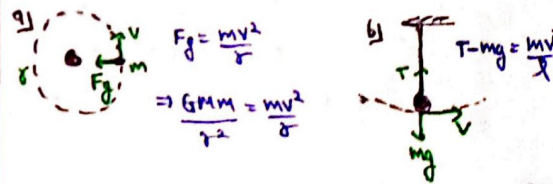
$$\Rightarrow a = \sqrt{a_t^2 + a_n^2} = \sqrt{(12)^2 + (162)^2} = 162.4 \text{ m/s}^2 \text{ Ans}$$

Centripetal Force:

Net force towards centre of circular path.

A force already in the system only starts serving as the centripetal force and enables circular motion.

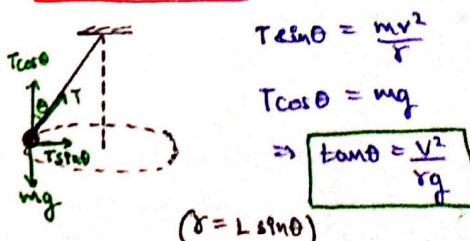
$$F_{net} = ma_n = \frac{mv^2}{R}$$



$$F_g = \frac{mv^2}{r} \Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$T = \frac{mv^2}{r} = m\omega^2 r$$

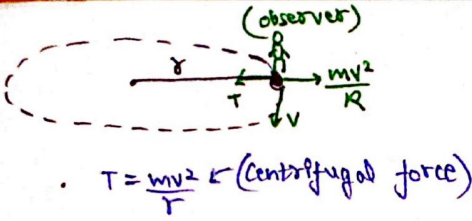
Conical Pendulum:



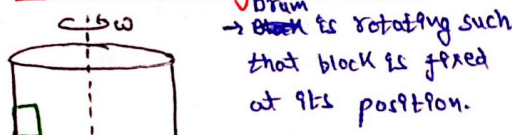
Centrifugal Force:

It is like circular pseudo force. Acts on body when particle/body is at rest w.r.t observer.

Acts opp. to centripetal force



Block in a Rotating hollow Drum:



For not slipping

$$f_{\max} > mg$$

$$\Rightarrow \mu_s N > mg \Rightarrow \mu_s \omega^2 MR > mg$$

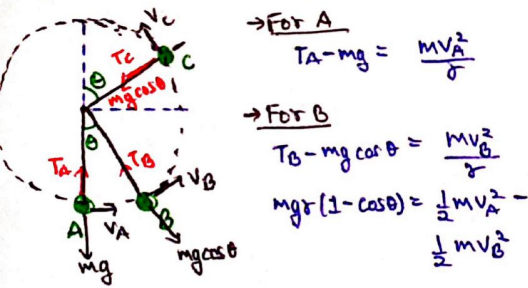
$$\Rightarrow \omega > \sqrt{\frac{g}{\mu_s R}}$$

i.e.

$$\omega_{\min} = \sqrt{\frac{g}{\mu_s R}}$$

Dynamics of Circular Motion:

→ Particle rotating along a circular path in vertical plane.



→ For C

$$T_C + mg \cos \theta = \frac{mv_C^2}{r}$$

$$mgr(1 + \cos \theta) = \frac{1}{2}mv_A^2 - \frac{1}{2}mv_C^2$$

Vertical Circular Motion of a Pendulum:

Case-01: $u \leq \sqrt{2gl}$:-

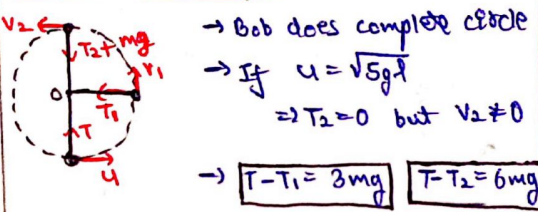
→ If $u = \sqrt{2gl}$, bob completes quarter circle ($\theta = 90^\circ$)

→ If $u < \sqrt{2gl}$, string slack before achieving $\theta = 90^\circ$ and oscillates to & fro.

Here,

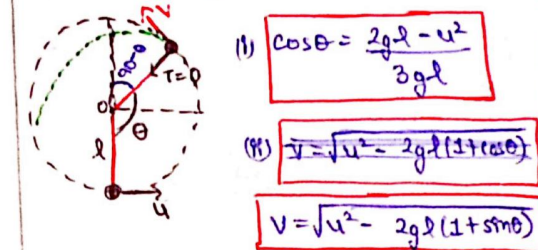
$$\cos \theta = \frac{2gl - u^2}{2gl}$$

Case-02: $u > \sqrt{5gl}$:-



Case-03: $\sqrt{2gl} < u < \sqrt{5gl}$:-

→ Here, bob rises above horizontal and somewhere string slacks ($T=0$) and bob does projectile motion.

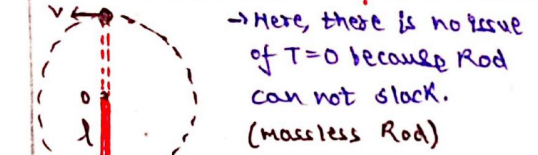


(iii)

$$H = \frac{v^2 \cos^2 \theta}{2g}$$

($\because \sin(90 - \theta) = \cos \theta$)

Vertical Circular Motion of Vertical Rod:

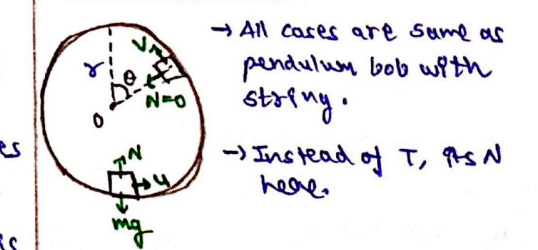


→ For $u > \sqrt{4gl}$

→ completes circle

→ Even for $u = \sqrt{4gl}$ → completes circle due to inertia ($v=0$)

Vertical Circular Motion Inside a Circular Track:



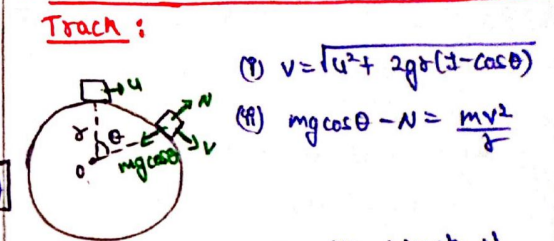
(i) $u > \sqrt{5gr} \Rightarrow$ completes circle

(ii) $\sqrt{2gr} < u < \sqrt{5gr} \Rightarrow$ cross horizontal at some θ with vertical $N=0$

$$\cos \theta = \frac{u^2 - 2gr}{3gr}$$

$$V = \sqrt{u^2 - 2gr(1 + \cos \theta)}$$

Vertical Circular Motion Outside a Circular Track:



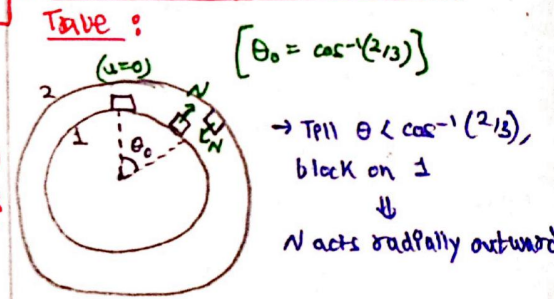
Block will lose contact with track at

$$\theta = \frac{u^2 + 2gr}{3gr}$$

If $u = 0 \Rightarrow \cos \theta = 2/3$

Without any initial velocity, block loose contact with track at this angle.

Vertical Circular Motion Inside a



→ Till $\theta > \cos^{-1}(2/3)$, block on 2

↓

N acts radially inward.

Banked Road Turning:

For no slipping $\Rightarrow$

$$\sqrt{Rg \left(\frac{\tan \theta - \mu}{1 + \mu \tan \theta} \right)} \leq v \leq \sqrt{Rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)}$$

3) If $v = \sqrt{Rg \tan \theta} \Rightarrow$

No friction would act on car as it can turn safely on slippery road.

3) If

$v > \sqrt{Rg \tan \theta} \Rightarrow f$ acts down the slope

$v < \sqrt{Rg \tan \theta} \Rightarrow f$ acts up the slope.